Problem 1:

$$\dot{x}_1(t) = \frac{-1}{x_2^2(t)}$$

$$\dot{x}_2(t) = u(t)x_1(t)$$

Since $\tilde{u}(t) = 0 \quad \forall t \geq 0$,

$$\dot{\tilde{x}}_2(t) = 0 \quad \forall t \geq 0$$

$$\Rightarrow \tilde{x}_2(t) = C$$

since $\tilde{x}_2(0) = 1$, $C = 1$, so $\tilde{x}_2(t) = 1$

Hence $\dot{\tilde{x}}_1(t) = -\frac{1}{1^2} = -1$

$$\Rightarrow \tilde{x}_1(t) = -t + A$$

$\tilde{x}_1(0) = 1$, so $1 = 0 + A \Rightarrow A = 1$

Hence $\tilde{x}_1(t) = -t + 1 = 1 - t$

$$\dot{x} = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{x_2^2(t)} \\ u(t)\, x_1(t) \end{bmatrix} = f(x, u)$$

To linearize, compute the Jacobian

$$A(t) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} \qquad -(x_2(t))^{-2}$$

$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & \frac{2}{x_2^3(t)} \\ u(t) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

plug in $\tilde{x}(t) = \begin{bmatrix} 1-t \\ 1 \end{bmatrix}$ and $\tilde{u}(t) = 0$



$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 & \frac{2}{x_2^3(t)} \\ u(t) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & \frac{2}{(1)^3} \\ 0 & 0 \end{bmatrix}$$

$$\left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

$$B(t) = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}} = \begin{bmatrix} 0 \\ x_1(t) \end{bmatrix}_{\substack{x=\tilde{x}(t)\\u=\tilde{u}(t)}}$$

$$\tilde{x}(t) = \begin{bmatrix} 1-t \\ 1 \end{bmatrix} \quad \text{so} \quad B(t) = \begin{bmatrix} 0 \\ 1-t \end{bmatrix}$$

Thus,

$$\delta\dot{x}(t) = A(t)\,\delta x(t) + B(t)\,\delta u(t)$$

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ 1-t \end{bmatrix} \delta u(t)$$

Problem 2:

State equations:

$$p(k+1) = p(k) + T_s\, v(k)$$

$$v(k+1) = v(k) + T_s\left(u(k) - \left(c\, v(k)\,|v(k)|\right)\right)$$

Output equation:

$$y(k) = v(k)$$

$$x(k) = \begin{pmatrix} p(k) \\ v(k) \end{pmatrix}$$

1) $x(k+1) = f(x(k), u(k))$

$y(k) = h(x(k), u(k))$

$$x(k+1) = \begin{pmatrix} p(k+1) \\ v(k+1) \end{pmatrix} = \begin{pmatrix} p(k) + T_s\, v(k) \\ v(k) - T_s\left(c\, v(k)\,|v(k)|\right) + T_s u(k) \end{pmatrix} = \begin{pmatrix} f_1(x(k), u(k)) \\ f_2(x(k), u(k)) \end{pmatrix}$$

$$= f(x(k), u(k))$$

$$y(k) = v(k) = h(x(k), u(k))$$

c) For diverse flow systems, eq point occurs when

Given that $v^* > 0$ is constant

Define: $x = \begin{pmatrix} p(k) \\ v(k) \end{pmatrix}$

a) $v(k) = v^* > 0$ (constant)

Need $v(k+1) = v(k) = v^*$ to maintain velocity.

Since $v^* > 0$, $|v^*| = v^*$

$$v(k+1) = v(k) - T_s C v(k)|v(k)| + T_s u(k)$$

Substitute v^*

$$v^* = v^* - T_s C v^{*2} + T_s u^*$$

$$T_s C v^{*2} = T_s u^*$$

$$\boxed{u^* = C(v^*)^2}$$

b) since $y(k) = v(k)$, y evaluated at a point yields:

$$\boxed{y^* = v^*}$$

c) nominal initial position: p_0^*

$$p^*(k) = p_0^* + kv^*$$

? need to multiply velocity by discrete timesteps since initial position. However, to account for sampling time, need to do kT_s.

thus,

$$\boxed{p^*(k) = p_0^* + kT_s v^*}$$

since $x_2 = v^*(k) = v^*$,

$$\boxed{x(k) = \begin{bmatrix} p(k) \\ v(k) \end{bmatrix} = \begin{bmatrix} p_0^* + kT_s v^* \\ v^* \end{bmatrix}}$$

3) Linearize around $(x^*(k), u^*)$

a) Define deviation variables:

$$\delta x(k) = x(k) - x^*(k)$$

$$\delta u(k) = u(k) - u^*$$

$$\delta y(k) = y(k) - y^*$$

b) Compute Jacobian Matrices:

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial p} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial p} & \frac{\partial f_2}{\partial v} \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$= \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s c \frac{\partial}{\partial v}\left(v(k)|v(k)|\right) \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$x^*(k) = \begin{bmatrix} p_0^* + k T_s v^* \\ v^* \end{bmatrix}$$

$$= \begin{bmatrix} 1 & T_s \\ 0 & 1 - T_s c \frac{\partial}{\partial v}\left(v|v|\right) \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

Since $v^* > 0$, evaluate $\frac{\partial}{\partial v}\left(v|v|\right)$ as if $v > 0$.

Hence, $\frac{\partial}{\partial v}\left(v|v|\right) = \frac{\partial}{\partial v}\left(v^2\right) = 2v(k)$

$$A = \begin{bmatrix} 1 & T_s \\ 0 & 1 - 2cT_s v^* \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}}$$

$$= \begin{bmatrix} 0 \\ T_s \end{bmatrix}_{\substack{x=x^*(k) \\ u=u^*}} = \begin{bmatrix} 0 \\ T_s \end{bmatrix} = B$$

c)

$$C = \left.\frac{\partial h}{\partial x}\right|_{\substack{x=x^*(k)\\u=u^*}} = \left[\frac{\partial h}{\partial p} \quad \frac{\partial h}{\partial v}\right]_{\substack{x=x^*(k)\\u=u^*}}$$

$$= \begin{bmatrix} 0 & 1 \end{bmatrix}_{\substack{x=x^*(k)\\u=u^*}}$$

$$\boxed{C = \begin{bmatrix} 0 & 1 \end{bmatrix}}$$

$$D = \left.\frac{\partial h}{\partial u}\right|_{\substack{x=x^*(k)\\u=u^*}} = \left[\frac{\partial h}{\partial u}\right]_{\substack{x=x^*(k)\\u=u^*}}$$

$$\boxed{D = [0]}$$

d)

$$\delta x(k+1) = \begin{bmatrix} 1 & T_s \\ 0 & 1 - 2cT_s v^* \end{bmatrix} \delta x(k) + \begin{bmatrix} 0 \\ T_s \end{bmatrix} \delta u(k)$$

$$\delta y(k) = \begin{bmatrix} 0 & 1 \end{bmatrix} \delta x(k) + [0]\,\delta u(k)$$

4) $T_s = 0.1\,s$

$c = 0.2$

$v^* = 10 \frac{m}{s}$

$$A = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 - 2(0.2)(0.1)(10) \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 - 0.4 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 1 & 0.1 \\ 0 & 0.6 \end{bmatrix}}$$

$$\boxed{B = \begin{bmatrix} 0 \\ T_s \end{bmatrix} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix}}$$

$$\boxed{C = [0 \quad 1]}$$

$$\boxed{D = [0]}$$

5)

a) The linearized model depends on v^* because the input required to maintain a velocity of v^* changes as v^* increases. Additionally, v^* changes how fast the position $p(k)$ of the vehicle updates. Thus, the linearized model depends on v^*.
The linearized model also depends on v^* due to the $v(k)|v(k)|$ term in $v(k+1)$. The method of calculating $\frac{\partial}{\partial v}(v|v|)$ at v^* changes based on the sign of v^*.

b) If the model was linearized at a different velocity, the only item that would change is the 2,2 entry of A.
Additionally, the input required to maintain v^* would also increase.

Problem 2.3 Txt:

Define $x_1(t) = \theta$, $x_2(t) = \dot{\theta}$, $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

We have:

$$ml^2\ddot{\theta} = mgl\sin\theta - b\dot{\theta} + T$$

$$T = \text{Sat}(u) = \begin{cases} -1 & u < -1 \\ u & -1 \le u \le 1 \\ 1 & u > 1 \end{cases}$$

$$y = \theta = x_1(t) = g(x, u)$$

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} \qquad \dot{x}_1(t) = \dot{\theta} = x_2(t)$$

$$\dot{x}_2(t) = \ddot{\theta} = \frac{mgl\sin\theta - b\dot{\theta} + T}{ml^2}$$

$$\dot{x}_2(t) = \frac{g}{l}\sin\theta - \frac{b}{ml^2}\dot{\theta} + \frac{T}{ml^2}$$

$$\dot{x}_2(t) = \frac{g}{l}\sin x_1(t) - \frac{b}{ml^2}x_2(t) + \frac{\text{Sat}(u)}{ml^2}$$

Thus,

$$\dot{x}(t) = \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} x_2(t) \\ \frac{g}{l}\sin x_1(t) - \frac{b}{ml^2} x_2(t) + \frac{sat(u)}{ml^2} \end{bmatrix} = f(x,u)$$

a) Given that eq. point exists where $\theta = x_1^* = 0$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{l}\sin x_1^* - \frac{b}{ml^2} x_2^* + \frac{sat(u^*)}{ml^2} \end{bmatrix}$$

Thus, $x_2^* = 0$.

$$\frac{g}{l}\sin \overset{0}{x_1^*} - \frac{b}{ml^2}\overset{0}{x_2^*} + \frac{sat(u^*)}{ml^2} = 0$$

$$\frac{sat(u^*)}{ml^2} = 0$$

$$sat(u^*) = 0$$

$sat(u^*) = 0$ when $u^* = 0$

now, $x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $u^* = 0$

Linearized Model:

$$\delta\dot{x}(t) = A\delta x(t) + B\delta u(t)$$

$$\delta y(t) = C\delta x(t) + D\delta u(t)$$

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

$$= \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(x_1) & -\frac{b}{ml^2} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

Substitute $x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $u^* = 0$:

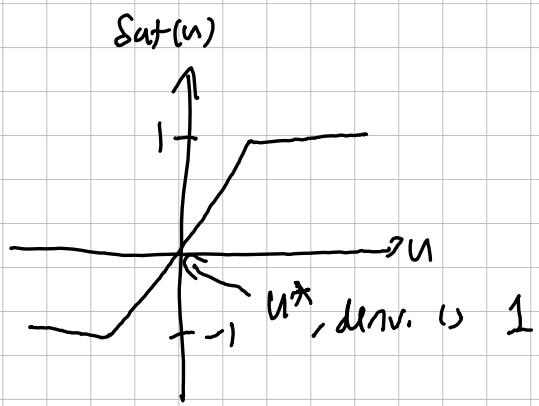
$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(0) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

$$= \begin{bmatrix} 0 \\ \frac{1}{me^2} \frac{\partial}{\partial u} sat(u) \end{bmatrix}_{\substack{x=x^* \\ u=u^*}}$$

Evaluate B at eq. point:

$$= \begin{bmatrix} 0 \\ \frac{1}{me^2} \left.\frac{\partial}{\partial u} sat(u)\right|_{u=u^*} \end{bmatrix}$$



$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix}$$

$$g(x,t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial u} \end{bmatrix} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\, \delta u(t)$$

b) given that eq. point exists where $\theta = x_1^* = \pi$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{\ell} \sin x_1^* - \frac{b}{m\ell^2} x_2^* + \frac{sat(u^*)}{m\ell^2} \end{bmatrix}$$

$$x_2^* = 0$$

$$0 = \frac{g}{\ell} \underbrace{\sin(x_1^*)}_{\sin(\pi)=0} - \frac{b}{m\ell^2} x_2^* + \frac{sat(u^*)}{m\ell^2}$$

$$0 = \frac{sat(u^*)}{m\ell^2} \Rightarrow sat(u^*) = 0$$

$$\Rightarrow u^* = 0$$

Hence $x^* = \begin{bmatrix} \pi \\ 0 \end{bmatrix}$ $u^* = 0$

Formulas for A, B, C, D stay same

but evaluate at new x^* and u^*

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(\pi) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no substitution required)}$$

$$g(x,t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial u} \end{bmatrix} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\, \delta u(t)$$

c) given that eq. point exists where $\theta = x_1^* = \frac{\pi}{4}$ Find x_2^* and u^*

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ \frac{g}{l}\sin x_1^* - \frac{b}{ml^2}x_2^* + \frac{sat(u^*)}{ml^2} \end{bmatrix}$$

$$x_2^* = 0$$

$$0 = \frac{g}{l}\sin(x_1^*) - \frac{b}{ml^2}\overset{0}{x_2^*} + \frac{sat(u^*)}{ml^2}$$

$$0 = \frac{g}{l}\frac{\sqrt{2}}{2} + \frac{sat(u^*)}{ml^2}$$

$$-\frac{g\sqrt{2}}{2l} = \frac{sat(u^*)}{ml^2}$$

$$\frac{-mgl^2\sqrt{2}}{2l} = sat(u^*)$$

$$-\frac{mgl\sqrt{2}}{2} = sat(u^*)$$

range of sat(u) function is

$[-1, 1]$. Therefore, eq point only

exists when

$$-\frac{mgl\sqrt{2}}{2} \in [-1, 1], \text{ so } u^* \text{ can}$$

be solved for.

For rest of problem, assume that

$-\frac{mgl\sqrt{2}}{2} \in [-1, 1]$ so system can be

linearized. Further, you could say that

$-\frac{mgl\sqrt{2}}{2} \in [-1, 0]$ since $mgl \geq 0$ (nonnegative quantities realistically)

Hence

$$-\frac{mgl\sqrt{2}}{2} = u^* \quad \text{(assume to proceed)}$$

Formulation of A, B, C, D matrices stay the same, only substitution of x^* and u^* changes.

$$x^* = \begin{bmatrix} \frac{\pi}{4} \\ 0 \end{bmatrix} \qquad u^* = \frac{-mgl\sqrt{2}}{2}$$

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos\left(\frac{\pi}{4}\right) & -\frac{b}{ml^2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{g\sqrt{2}}{2l} & -\frac{b}{ml^2} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no switching required)}$$

$$g(x,t) = x_1(t) = \theta$$

$$C = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial x_1} & \frac{\partial g}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=x^* \\ u=u^*}} = \begin{bmatrix} \frac{\partial g}{\partial u} \end{bmatrix} = [0]$$

Final

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g\sqrt{2}}{2l} & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\,\delta u(t)$$

d) $b = \frac{1}{2}$ $\quad mgl = \frac{1}{4}$ compute

$T(t)$ needed so that pendulum follows

from $\theta(0) = 0$ w/ $\dot{\theta}(t) = 1 \quad \forall t \geq 0$

since $\dot{\theta}(t) = 1 \quad \forall t \geq 0, \quad \ddot{\theta}(t) = 0 \quad \forall t \geq 0$

$$ml^2\ddot{\theta} = mgl\sin\theta - b\dot{\theta} + T$$

$$0 = mgl\sin\theta - b + T$$

$$\tilde{T}(t) = b - mgl\sin\theta(t)$$

$$\tilde{T}(t) = \frac{1}{2} - \frac{1}{4}\sin\theta(t)$$

$$\dot{\tilde{x}}_1(t) = \dot{\tilde{\theta}}(t) = 1 \quad \forall t \geq 0$$

$$\tilde{x}_1(t) = t + C, \text{ since } x_1(0) = 0, \quad C = 0$$

$$\tilde{x}_1(t) = t$$

Now, $\tilde{T}(t) = \frac{1}{2} - \frac{1}{4}\sin t$

$$\dot{\tilde{x}}_2(t) = \ddot{\tilde{\theta}}(t) = 0 \Rightarrow \tilde{x}_2(t) = \dot{\tilde{\theta}}(t) = 1 \quad \forall t \geq 0$$

Hence

$$\tilde{x}_1(t) = t$$

$$\tilde{x}_2(t) = 1$$

$$\tilde{T}(t) = \frac{s}{2} - \frac{1}{4}\sin t$$

Linearize around this trajectory.

The A, B, C, D matrices formula stay the same, but substitute time varying trajectories instead of equilibrium point ⟶

$$A(t) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(x_1) & -\frac{b}{ml^2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=0(t)}}$$

$$A = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(t) & -\frac{b}{ml^2} \end{bmatrix}$$

$$B(t) = \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \quad \text{(no saturation required)}$$

$$y(x,t) = x_1(t) = \theta$$

$$C(t) = \left.\frac{\partial g}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \left[\frac{\partial g}{\partial x_1} \quad \frac{\partial g}{\partial x_2}\right]_{\substack{x=\tilde{x}(t) \\ u=0(t)}} = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D(t) = \left.\frac{\partial g}{\partial u}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \left[\frac{\partial g}{\partial u}\right]_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = [0]$$

Hence:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ \frac{g}{l}\cos(t) & -\frac{b}{ml^2} \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ \frac{1}{ml^2} \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\,\delta u(t)$$

Problem 2.4 Text

$$\ddot{\theta} + k\sin\theta = \tau$$

a) $u := \tau$ is input

$y := \theta$ is output.

Let $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$

Then $\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$

$\dot{x}_1 = \dot{\theta} = x_2$

$\dot{x}_2 = \ddot{\theta} = \tau - k\sin\theta = -k\sin x_1 + u$

$y = \theta = x_1$

Hence

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -k\sin x_1 + u \end{bmatrix} = f(x,u)$$

$$y = x_1 = g(x,u)$$

b) Find eq. points of system corresp to $z(t) = 0 \quad \forall t \geq 0$

At eq,

$f(x, u) = \vec{0}$. Hence:

$$\dot{x}^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x_2^* \\ -k \sin x_1^* + u^* \end{bmatrix}$$

$u^* = z^* = 0$ (given)

$x_2^* = 0$ by observation

$-k \sin x_1^* = 0$

$\sin x_1^* = 0$

$x_1^* = 0 + \pi k, \quad k \in \mathbb{Z}$

Hence:

$$x^* = \begin{bmatrix} 0 + \pi k \\ 0 \end{bmatrix}, \quad k \in \mathbb{Z}, \quad u^* = 0$$

c) Linearize system around solution

$$\tau(t) = \theta(t) = \dot{\theta}(t) = 0 \quad \forall t \geq 0$$

Treat given as trajectories:

$$\tilde{u}(t) = \tilde{\tau}(t) = 0 \qquad \forall t \geq 0$$

$$\tilde{x}_1(t) = \theta(t) = 0$$

$$\tilde{x}_2(t) = \dot{\theta}(t) = 0$$

Hence:

$$A = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$= \begin{bmatrix} 0 & 1 \\ -k\cos(x_1) & 0 \end{bmatrix}_{\substack{x=\tilde{x}(t) \\ u=\tilde{u}(t)}}$$

$$\tilde{x}(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \tilde{u}(t) = 0, \text{ so } A = \begin{bmatrix} 0 & 1 \\ -k\cos(0) & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 \\ -k & 0 \end{bmatrix}$$

$$B = \left.\frac{\partial f}{\partial u}\right|_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}} = \begin{bmatrix} \frac{\partial f_1}{\partial u} \\ \frac{\partial f_2}{\partial u} \end{bmatrix}_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \left.\frac{\partial y}{\partial x}\right|_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} & \frac{\partial y}{\partial x_2} \end{bmatrix}_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix}_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

$$D = \left.\frac{\partial y}{\partial u}\right|_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}} = \left[\frac{\partial y}{\partial u}\right]_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}} = [0]_{\substack{x=\hat{x}(t)\\u=\hat{u}(t)}}$$

$$D = [0]$$

Hence, linearized model is:

$$\delta\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -k & 0 \end{bmatrix} \delta x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \delta u(t)$$

$$\delta y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \delta x(t) + [0]\, \delta u(t)$$